



Internship Report

Transport Analytics & Operations Intelligence
for the Wasla Platform

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Degree program	Master's Degree in Statistics and Data Science
University	University of Palermo (Italy)
Host company	Backend Glitch Inc. (backendglitch.com)
Company supervisor	Mr. Samer Gassouma (CEO)
Location	Monastir, Tunisia
Period	1 July 2026 to 25 September 2026



Project repository



Live platform

1 Host institution

Backend Glitch Inc. (<https://backendglitch.com>) is a software company based in Monastir, Tunisia. The company builds software solutions for companies and for the clients who come to it: it works mainly on fintech and crypto solutions and on backend solutions, and it also develops its own products, some of which are still on the way. One of these projects is *Wasla*, a transport management platform that the company built as a solution for its client **Dhraief Service de Transport** (<https://www.dhraief.live/>), the operator of the Monastir Centre and Jammel Monastir stations.



Dhraief Service de Transport, the client behind the Wasla platform

Wasla is in production at both stations, with a third station (Ksar Hlel) planned. On a normal day the platform handles everything that happens at a station: passenger bookings, vehicle dispatching, day passes, route scheduling, ticket printing and staff shift tracking. The fleet counts 113 vehicles of eight seats each, serving five destinations. The system has been collecting data since April 2026, and by the time I joined there was more than a year's worth of bookings, trips and sales sitting in the databases with nobody looking at it analytically. That was the whole point of my internship.

2 Dates and location

I did my internship in Monastir, Tunisia, from 1 July 2026 to 25 September 2026, working in person at the company under the supervision of Mr. Samer Gassouma, the CEO. Most of my time was spent in the office and on the station servers, which are only reachable from inside the local network.

3 Objectives of the approved project

The training plan agreed with the university listed six objectives:

1. Understand the architecture and operational workflow of the Wasla platform.
2. Analyse booking data, revenue streams and fleet utilisation statistically.
3. Develop time-series models to forecast passenger volume.
4. Design anomaly detection for booking patterns.
5. Build analytical dashboards for operations monitoring and KPIs.
6. Deploy the analytical services in a cloud environment (Docker, Linux VPS).

4 Work actually performed

The work naturally fell into five phases, and this is roughly how my three months went.

Phase 1: getting the data out

The first thing I learned is that real production data does not wait for you in a CSV file. The two stations each run their own PostgreSQL database on their own server, accessible only over SSH tunnels, and for good reasons I was given read-only access. I wrote an extraction pipeline that pulls eight tables per station (bookings, trips, day passes, routes, vehicles, staff, print jobs and staff transactions) into parquet files, then a cleaning layer that merges the two stations into one dataset. This was more delicate than it sounds: the two databases do not even agree on column names (seats are `booked_seats` in one and `seats_booked` in the other), and each station uses its own destination IDs. I built a mapping between the two namespaces so the five routes become comparable across stations. In the end the cleaned dataset holds **1,582,729 bookings** from October 2025 to October 2026, plus 15,470 trips and 39,792 day passes.

Two discoveries from this phase shaped everything that followed. First, roughly 93% of the bookings in the system are *ghost bookings*: pre-generated drafts that were never actually sold. Only about 6.6% are verified tickets. Second, the word “revenue” means two different things in this business. What passengers pay (`total_amount`, which includes the driver’s base fare) is one number; what the station actually keeps (0.15 TND per seat plus 2 TND per day pass) is another, about fourteen times smaller. I decided from the start that every chart, table and KPI would say explicitly which revenue it is showing, because mixing the two makes the numbers look wrong even when they are both right.

A closer look: how the data went from database to chart

Since the pipeline is the part of my work I was asked about the most, here it is step by step, with the actual commands I used every morning.

Step 1. Open the tunnels. Each station server is reached through a local port forward, so the databases stay private while my scripts see them as local Postgres instances.

```
ssh -fN -L 15432:localhost:5432 server@100.72.205.59
ssh -fN -L 15433:localhost:5432 ste@100.123.86.114
```

Step 2. Extract. One script per station pulls every table and stores it as parquet, batching large tables by id so a million-row bookings table never fills the memory.

```
cd Phase_1/scripts
python3 01_extract.py jemmal
python3 01_extract.py monastir
```

Step 3. Clean and enrich. A cleaning script unifies the two schemas, then a features script adds the time variables (hour, weekday, month) and the real versus ghost flag that every downstream analysis relies on.

```
cd .. && python3 02_clean.py && python3 03_features.py
```

Step 4. Plot and analyse. The Phase 2 scripts group the enriched bookings by hour, by day, by route and by vehicle and write both the numbers and the charts. For example, the hourly flow chart below is exactly what came out of this step for the whole history, and it is where the two peaks became visible.

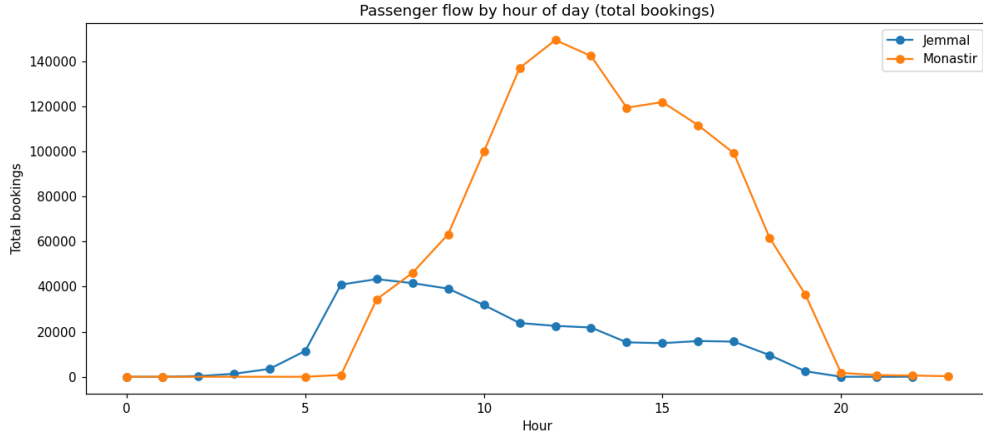


Figure 1: Passenger flow by hour of day, both stations. Jemmal peaks at 07h and Monastir at 12h.

Step 5. Forecast and detect anomalies. Phase 3 turns the daily series into training data, runs the walk-forward cross-validation, refits the winning model on the full series and produces the seven-day forecast with its confidence bands, then scores every day with the three anomaly detectors.

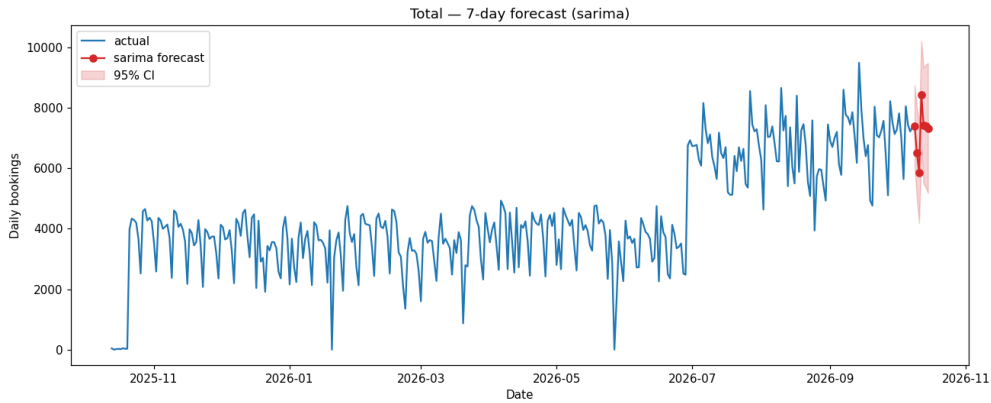


Figure 2: Consolidated forecast: the SARIMA model with its walk-forward validation and the seven-day horizon ahead of the last training day.

Step 6. Serve. The dashboard reads these same artifacts, so whatever the screen shows can always be traced back to a file, and the API exposes the same numbers over HTTP with a live feed on top.

Phase 2: statistics

With the data in shape I analysed passenger flow, revenue, routes and fleet use. The clearest results: Jemmal peaks at 07h and Monastir at 12h (both around three times their off-peak level), and Jemmal's fleet runs at a 90.9% average load factor over 80 trips per vehicle. On the revenue side, the consolidated picture is 3,427,079 TND passenger-paid against 237,729 TND of station fees. The route table showed something the operators already suspected: the Jemmal to Tunis line is the premium route, with an average ticket of 15.85 TND and almost no ghost bookings, while most other routes are dominated by drafts.

Phase 3: forecasting and anomaly detection

I built daily demand series per station and for the network, then tested six forecasting models: naive, seasonal naive, seasonal median, weekly mean, SARIMA and Prophet. Instead of a single train/test split I used walk-forward cross-validation, which feels like the only honest way to evaluate models on a time series: the model always predicts the week after everything it has seen. Prophet lost on every series; the winners were the simplest ones. Seasonal naive won for Monastir (MAPE 8.5%), naive won for Jemmal (6.3%), and SARIMA won for the consolidated total (6.0%). For anomaly detection I combined three detectors, a rolling z-score, a week over week difference z-score, and an Isolation Forest, and flagged a day when at least two of them agree. This combination caught things a single detector missed, like the two zero-booking days at Monastir that a plain z-score would never see.

Phases 4 and 5: the dashboard, the API and the deployment

The last stretch of the internship was software work. I built a seven-view analytics dashboard (overview, passenger flow, revenue, routes, fleet, forecast, anomalies) with Next.js and Recharts, designed so that every number on screen can be traced back to the exact file that produced it. Behind it sits a small FastAPI service that serves the analysis results and pushes a live minute-by-minute feed of today's bookings over a WebSocket. Both are packaged with Docker. The dashboard is deployed on Vercel at <https://wasla-analytics.vercel.app> (in static-data mode, since the station databases live behind the local network), and the whole codebase, including the data, is on GitHub at https://github.com/BackendGlitch/wasla_Analytics.

I also fixed my share of bugs along the way, the kind you only meet in production: a partial extraction day that skewed the MAPE to absurd values, a WebSocket handshake that FastAPI silently rejected because of a missing type annotation, and a half-terabyte CSV that would not fit on GitHub until it was compressed. Each one taught me something about verifying before trusting.

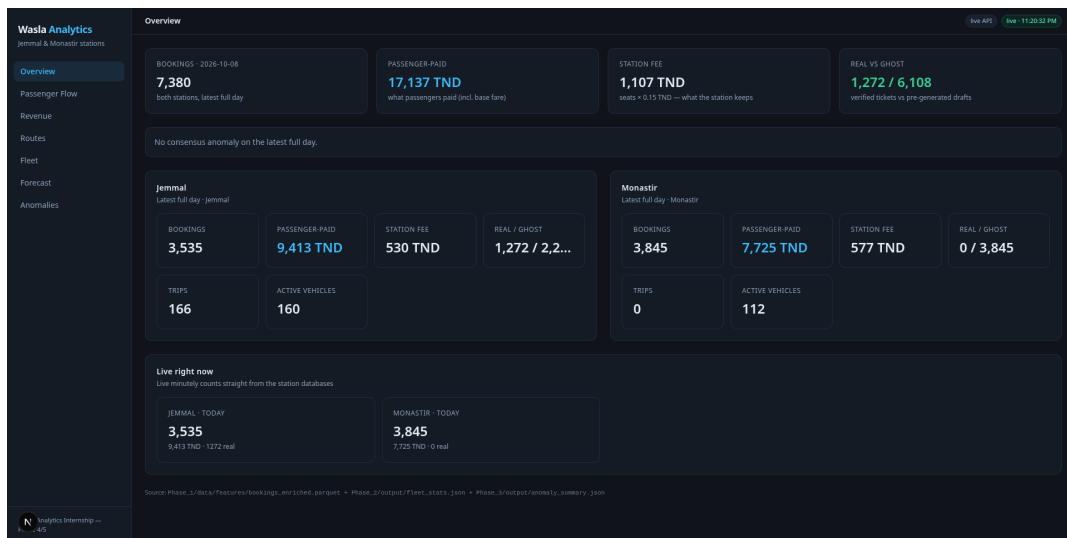


Figure 3: The Overview view of the dashboard, showing both revenue definitions and the live feed from the two stations.

5 Technologies and methods used

On the data side I worked with PostgreSQL over SSH tunnels, pandas and pyarrow for processing, and parquet for storage. The statistical work used scipy and statsmodels (descriptive analysis,

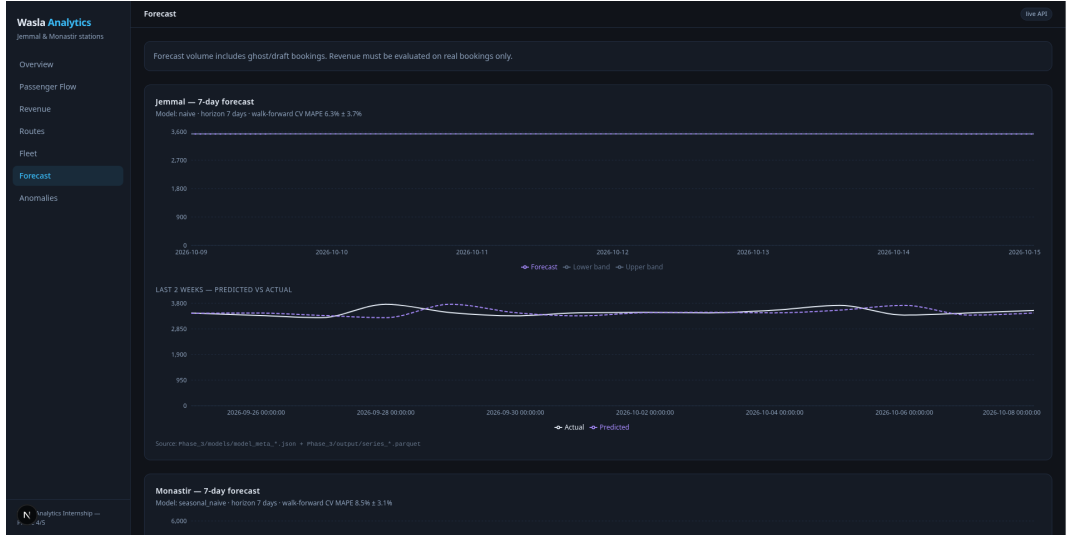


Figure 4: Forecast view: seven-day forecasts with confidence bands and the cross-validation MAPE for each model.

time-of-day and weekday profiles, load factors), and the machine-learning side used scikit-learn together with statsmodels for SARIMA and Prophet. Model evaluation was always walk-forward cross-validation with MAPE and RMSE, reported with their standard deviations across folds. The software side was Next.js with TypeScript, Tailwind and Recharts for the dashboard, FastAPI with WebSockets for the API, Docker and docker compose for packaging, and Git/GitHub with Vercel for deployment. One habit I kept for the whole internship: whenever a number looked surprising, I recomputed it from the raw parquet before touching any code. More than once the bug was upstream.

6 Results and outcomes

In numbers: the pipeline now produces a clean, unified dataset of 1.58 million bookings; the statistical reports quantify the two peaks (07h at Jemmal, 12h at Monastir), the two revenue definitions (3.43M TND passenger-paid versus 238k TND station fees), the five-route performance table and a 90.9% fleet load factor; the forecasting models deliver seven-day predictions with a cross-validated MAPE between 6.0% and 8.5%, served with confidence bands; and the anomaly system flagged a booking spike at Jemmal on 29 June 2026 and two zero-booking days at Monastir. All of it is visible in the dashboard, through the API, and reproducible from the GitHub repository. For the company, the most actionable outcomes are probably the ghost booking reality (revenue decisions should be made on verified tickets only) and the morning peak, which leaves almost no slack in the fleet.

7 How it relates to my Master’s studies

My degree in Statistics and Data Science at the University of Palermo prepared me for a large part of this work, and the internship returned the favour by making all of it concrete. The cross-validated model selection I used for the forecasts comes straight from the time-series courses, but applying it to data where one station only has four months of history, and where a single partial day can send the MAPE to 3000%, is a different experience from the textbook datasets. The anomaly detection work (multiple detectors, consensus voting, handling zero-inflated series) is applied statistical learning in the truest sense. And the deployment part, containers, a REST API, a WebSocket feed, a public

deployment, matches the cloud-computing component of the programme. What the courses could not teach, the company did: working with read-only production access, respecting a live system, and verifying every result against its source before presenting it.

8 Conclusion

Looking back at the three months, the internship turned out to be a full pipeline from database to dashboard, and I am glad it did: nothing less would have taught me how every stage depends on the one before it. I learned to deal with messy, inconsistent, real-world data; I built forecasting and anomaly-detection systems that survived cross-validation rather than just fitting; and I shipped software that people at the stations can actually open. The two findings that matter most for the company, the ghost-booking reality and the dual revenue definitions, are now measured, labelled and monitored instead of hidden in the database. I finished the internship with a much clearer idea of what it means to do data science on a live system, and I am grateful to Mr. Gassouma and the whole team for the opportunity.